

Teacher Code :

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MATHEMATICAL METHODS UNIT 2

Tech-Active

Test theme: THE HAIL MARY MANOEUVRE

Reading Time: 5 minutes

Writing time: 45 minutes

QUESTION AND ANSWER BOOK

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
3	3	30

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.
- Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your **name** in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer **all** questions in the spaces provided.

Unless otherwise specified an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

**Question 1** (12 marks)

Ryland Grace is orbiting the planet Adrian to collect Astrophage. To do this, the Hail Mary must dip into the planet's rings. The ship's altitude h (in kilometres) above the ring's median line is modeled by:

$$f(t) = A \sin(n(t - c)) + k, \quad t \in [0, 40]$$

Where t is the time in minutes since the orbit began.

The ship reaches a maximum altitude of $\sqrt{150} + 12$ km and a minimum altitude of $12 - \sqrt{150}$ km. It takes exactly $\frac{13\pi}{4}$ minutes to travel from a peak altitude to the following trough. The ship is at its average altitude at $t = 0$ and is currently moving upwards.

- a. Show that $A = 5\sqrt{6}$ and $k = 12$

2 marks

- b. Determine the exact values of n and the smallest non-negative value of c .

2 Marks

- c. Grace is running low of fuel. He can only harvest Astrophage when his altitude is below $12 - \frac{5\sqrt{6}}{2}$ km. Find the total amount of time, in minutes, that Grace is successfully harvesting Astrophage during the first 40 minutes of the mission. Give your answer correct to two decimal places.

3 Marks



- d. Grace meets a new friend, Rocky. To sync their orbits, Rocky tells Grace he must change his flight path. The new path $g(t)$ is a transformation of $f(t)$. Grace must:
1. Dilate the path by a factor of $\frac{1}{2}$ from the t -axis
 2. Reflect in the h -axis
 3. Translate it $\frac{\pi}{4}$ units in the positive t direction.

State the new equation $g(t)$ in the form $g(t) = a\sin(b(t - d)) + k$

3 Marks

e. Find the first time $t > 0$ where two flight paths $f(t)$ and $g(t)$ intersect. Give your answer to three decimal places

2 Marks

Question 2 (10 Marks)

Grace and Rocky are trying to find a specific Taumoaba virus located on a rotating asteroid.

The asteroid is perfectly circular with a radius of $\sqrt{108}$ meters. The asteroid rotates anti-clockwise, completing one full revolution every 18 minutes.

- a. Grace tethers himself to the edge of the asteroid at an initial position P defined by the angle $\theta = \frac{13\pi}{12}$ from the positive x -axis. Find the exact coordinates (x, y) of Grace's starting position.

2 Marks

- b. Find Grace's coordinates after 7 minutes of rotation. Give your answer in exact surd form

2 Marks

- c. A pocket of the virus is located at a fixed coordinate $V(-3, 3\sqrt{3})$. As Grace rotates, he uses a sensor to measure his straight-line distance D from the virus. Write an expression for $D(t)$, where t is the time in minutes

3 Marks



- d. Grace's trajectory goes wrong! An incoming asteroid fragment forces Grace to detach when his distance from the virus is exactly 10 metres. Find the first time $t > 0$ when this occurs. Give your answer in minutes, correct to two decimal places.

3 Marks



Question 3 (8 Marks)

To save Earth, Grace must send a video message back using a high-powered narrow-beam laser. However, there is an asteroid field in the way. The density of the asteroids $S(x)$ along the transmission path x is modeled by:

$$S(x) = \frac{1}{\sqrt{3}} \tan\left(\frac{\pi x}{20} - \frac{\pi}{6}\right) + \frac{2}{3}, \quad x \in [0, 25]$$

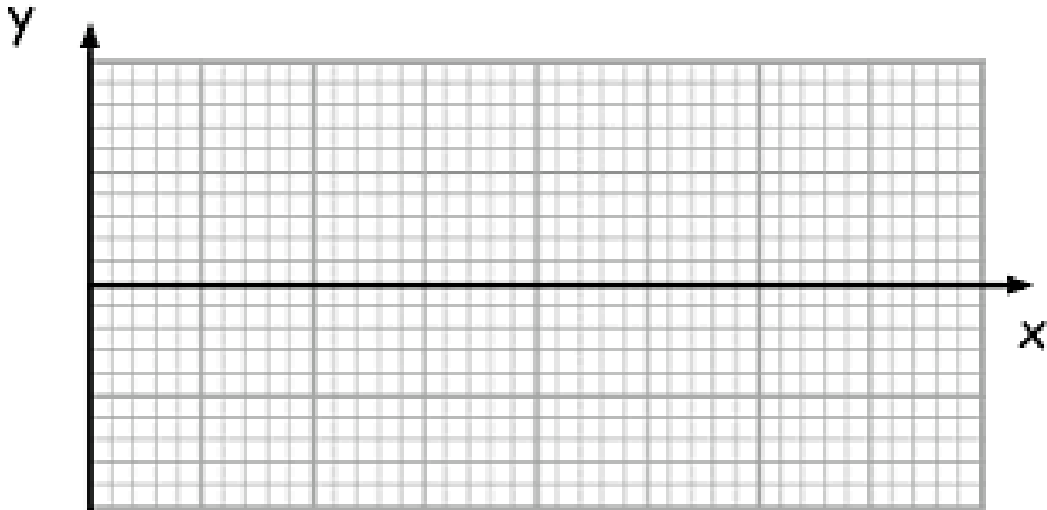
Where x is the distance from the ship in millions of kilometers.

- a. The video signal is completely blocked if it hits a ‘Gravitational Singularity,’ which occurs at the vertical asymptotes of $S(x)$. Find the distance x where the first singularity occurs.

2 Marks

- b. Sketch the graph of $y = S(x)$ for $x \in [0, 25]$. Label the axial intercepts and endpoints with their coordinates (rounded to 2 decimal places where not exact) and any asymptotes with their equations.

3 Marks



- c. The message is 'Safe to Read' only if the asteroid density is low, specifically when $0 < S(x) < 1$. Find the intervals of x for which the message is safe to read. Give your answer correct to three decimal places.

3 Marks

THE PETROVA LINE IS GLOWING. THE FATE OF TWO WORLDS IS IN YOUR

HANDS

END OF TEST... UNTIL THE NEXT REVOLUTION...